Check for updates

Received: 15 January 2024 | Accepted: 26 November 2024

DOI: 10.1111/bjop.12755

the british psychological society
promoting excellence in psychology

ARTICLE

Bridging minds and machines: Unmasking the limits in text-based automatic personality recognition for enhanced psychology–AI synergy

Avanti Bhandarkar[1] | **Ronald Wilson**[1] | **Anushka Swarup**[1] | **Gregory D. Webster**[2] | **Damon Woodard**[1]

[1]Florida Institute for National Security (FINS), Gainesville, Florida, USA

[2]Department of Psychology, University of Florida, Gainesville, Florida, USA

Correspondence
Avanti Bhandarkar, Florida Institute for National Security (FINS), 601 Gale Lemerand Dr., Gainesville, FL 32603, USA.
Email: avantibhandarkar@ufl.edu

Abstract

Text-based automatic personality recognition (APR) operates at the intersection of artificial intelligence (AI) and psychology to determine the personality of an individual from their text sample. This covert form of personality assessment is key for a variety of online applications that contribute to individual convenience and well-being such as that of chatbots and personal assistants. Despite the availability of good quality data utilizing state-of-the-art AI methods, the reported performance of these recognition systems remains below expectations in comparable areas. Consequently, this work investigates and identifies the source of this performance limit and attributes it to the flawed assumptions of text-based APR. These insights are obtained via a large-scale comprehensive benchmark and analysis of text data from five corpora with diverse characteristics and complementary personality models (Big Five and Dark Triad) applied to an assortment of AI methods ranging from hand-crafted linguistic features to data-driven transformers. Finally, the work concludes by identifying the open problems that can help navigate the limitations in text-based automatic personality recognition to a great extent.

KEYWORDS

automatic personality recognition, big five, dark triad, human–AI teaming, lexical hypothesis

© 2024 The British Psychological Society.

INTRODUCTION

Personality traits reflect an individual's distinct pattern of thoughts, emotions and actions that remain consistent over time (Goldberg, 1993). The modern trait theory posits that these traits are fundamental and individual behaviour could be grouped based on the strength of expression of these traits (Allport, 1937; Cattell, 1950; Eysenck, 2013). Further, the theory also classifies these traits in the context of social desirability, eventually resulting in some traits labelled as desirable and others as undesirable. However, in the age of artificial intelligence (AI), the role of personality traits as a mere tool for modelling individual behaviour has long evolved.

Personality modelling and recognition is widely used in many human-centric applications of AI to improve human–machine relationships. Personal digital assistants are made to be sensitive to the user's personality to understand, predict and cater to the user's needs. The customer service industry relies on chatbots to handle customer queries while being sensitive to their emotions and personality to ensure customer well-being and satisfaction. In a similar vein, personality profiling algorithms are employed by corporate companies to select prospective candidates that best fit the team and culture to ensure optimal productivity and success. With several online platforms providing ample opportunities for individuals to disclose their thoughts, feelings, emotions and opinions, there is a copious amount of user-generated data available to enable seamless automated personality detection and recognition via AI, especially from text data.

With text data being a treasure trove of information, several fields have actively pursued the development of methods to refine and utilize information from text data effectively. For instance, computational linguistics can effectively summarize information, extract relationships and entities present in text, and even respond to queries with near-accurate answers. With AI and its data-driven methodologies, these areas have grown substantially resulting in large-language models such as ChatGPT that can answer a multitude of questions in a variety of areas while even providing human comprehensible reasons. With strong theoretical foundations, namely the *Lexical Hypothesis*, supporting the use of language in assessing personality, automated personality recognition (APR) leverages the developments in AI and natural language processing (NLP) to evaluate personality from text. However, unlike other areas that experienced tremendous success, APR has remained close to the pre-AI era despite the availability of large quantities of diverse data and the utilization of advanced learning methods. Consequently, *the goal of this article was to investigate the source of the limitation preventing the effective utilization of AI for APR*. To this end, this article discusses the diversity of AI methods used in APR, and their utility, and sheds light on the issues limiting the growth of AI in personality psychology by discussing open problems to motivate both psychology and AI researchers.

The rest of the paper is structured as follows: Section 2 underscores the role of AI and the assumptions of APR. Section 3 provides a detailed breakdown of the data available for APR and subsequent AI methods developed in existing literature along with a comprehensive benchmark on the operational infrastructure revealing limitations due to the underlying assumptions of APR. Next, Section 4 quantitatively evaluates the limitations of APR and offers possible solutions via open problems in Section 5. Finally, the work is concluded in Section 7.

The key contributions of this article are as follows:

- A first-of-its-kind comprehensive evaluation of diverse data-driven text-based APR methods applied to text corpora of varying data characteristics (such as topics, genre and domain) and collection methodologies (controlled and uncontrolled) with respect to both desirable and undesirable personality traits represented by the Big Five and Dark Triad personality models, respectively.
- Empirical findings suggest that the theoretical foundation supporting the extraction of personality traits from text has practical limitations that detrimentally affect the use of AI in APR systems.
- Avenues for contribution from both psychology and AI experts via open problems to help address this limitation and make APR an effective and commonly available tool for widespread adoption.

20448295, 0, Downloaded from https://bpspsychub.onlinelibrary.wiley.com/doi/10.1111/bjop.12755 by University Of Arts London, Wiley Online Library on [21/01/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

BACKGROUND AND MOTIVATION

Language serves as the primary means by which people express their internal thoughts and emotions, making it fundamental to psychology and communication (Goldberg, 1981). It is through words and language that cognitive, personality, clinical and social psychologists seek to understand human behaviour (Tausczik & Pennebaker, 2009). Language is also central to human experience for cultural, social and biological reasons (John et al., 1988). In everyday life, people create words to describe both desirable and undesirable behaviours within their society. Over time, these words, known as lexical descriptors, become embedded in the community's vernacular. This process of everyday language evolving into lexical descriptors is central to the Lexical Hypothesis, which links language to personality (Goldberg, 1993). An empirical relationship among these descriptors forms the basis for assessing and measuring personality. Thus, language is regarded as a *sedimentary deposit*—a cumulative record of the observations of personality made by individuals who have used and shaped it throughout its history (Widiger & McCabe, 2020).

For example, at the language level, the personality trait of extroversion might be described by lexical descriptors such as 'assertive', 'friendly' and 'outgoing'. Similarly, this trait may manifest in everyday language through the use of positive and social words or phrases such as 'party', 'hang out' or 'catch up'. However, while the Lexical Hypothesis focusses on how everyday language evolves into lexical descriptors, the study of how these descriptors relate to actual language expression is less frequently explored in psychology theory, except in studies employing computerized text analysis methods such as LIWC. This aspect forms the crux of APR.

While the Lexical Hypothesis supports the assessment of personality through a forward process where lexical descriptors stem from everyday language use, APR operates in reverse. *If descriptive terms for assessing personality traits originate from common language use, then it stands to reason that everyday language also mirrors personality traits*. This interaction highlights a duality between the Lexical Hypothesis and APR. For example, individuals described as 'assertive' or 'friendly' might not often use these specific terms in daily conversation, yet these traits can be inferred from their language patterns and expressions. Moreover, while conceptual trait descriptors tend to retain their meaning over time, the language people use continually evolves. Consider the recent emergence of internet slang such as 'rizz', which now colloquially encapsulates qualities traditionally described as 'charming' or 'attractive'. This illustrates how changes in everyday language can lead to new lexical descriptors. Despite these changes, a comprehensive theory explaining how this evolution influences personality assessment through everyday language use is still lacking.

Assumptions in APR

The advent of online platforms has transformed how people share thoughts, emotions and opinions, significantly increasing the volume of data available for personality analysis. This surge in digital communication, combined with the advancement of AI technologies such as sophisticated machine learning (ML) and deep learning (DL) methods, has greatly enhanced the capacity for conducting covert personality assessments through text analysis. Despite the availability of state-of-the-art (SOTA) AI methods that facilitate APR, the fundamental assumption underpinning this approach is that *personality traits are indeed reflected in text*. If this assumption does not hold, the foundation for profiling personality from text is called into question.

In practice, APR methods analyse language at the population level by training on text data from numerous individuals. These methods use personality labels derived from self-reported questionnaires, often constructed based on the Lexical Hypothesis such as the Big Five Inventory (BFI), IPIP-NEO or Dirty Dozen (Goldberg et al., 1999; John et al., 1991; Jonason & Webster, 2010). This creates an implicit assumption in APR that there is a relationship between the lexical descriptors used in these questionnaires, the assigned personality scores and the text data. Traditional APR approaches, such as those using

20448295, 0, Downloaded from https://bpspsychub.onlinelibrary.wiley.com/doi/10.1111/bjop.12755 by University Of Arts London, Wiley Online Library on [21/01/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

tools such as LIWC, operate under the assumption that specific word use reflects personality traits, which we refer to as trait lexical markers (Yarkoni, 2010). For instance in the above examples, the positive connotation words 'met', 'friends', 'great time' might be reflective of the individual's high level of extroversion as opposed to the negative connotation words such as 'dragged' and 'waste of time' suggesting otherwise.

However, large-scale studies in psychology investigating the correlation between lexical markers and personality traits often reveal significant yet weak correlations (Koutsoumpis et al., 2022; Moreno et al., 2021; Preotiuc-Pietro et al., 2016; Sumner et al., 2012). Similarly, despite the availability of large amounts of data and sophisticated AI techniques, methods reported in AI literature fail to meet the expected performance (Feizi-Derakhshi et al., 2022). Take, for example, another comparable aspect of psychology that measures affective states—sentiment realized through NLP in the form of sentiment analysis (SA). SA is very similar to APR in terms of the availability of data with the same degree of complex AI methods available in its associated literature. However, SA achieves near-perfect detection rates of sentiments from text, likely because sentiment labels are often derived from self-reported experience ratings directly linked to specific texts, such as Yelp or IMDB reviews. This direct association between sentiment labels and text contrasts sharply with APR, where no such direct link exists between the personality trait scores from questionnaires and the specific text data analysed. This discrepancy suggests that linguistic attributes reflecting personality may not be readily available in the text—necessitating the reevaluation of the foundational assumptions of APR.

From a text analysis perspective, accurately modelling a characteristic hinges on two key properties: *perceptibility* and *ubiquity*. Consider the example text: 'My father was a strict and authoritative figure in my growing up years. He always reprimanded me for the smallest of mistakes'. While it may seem challenging to assess the personality of someone based solely on their description of another person, the narrative could imply the individual's resentment towards their father, potentially signalling lower agreeableness. However, drawing a definitive personality assessment from this text alone is difficult without more context, highlighting that *personality traits may not always be perceptible in the texts analysed*. Moreover, even if language use generally reflects personality, the specific lexical markers indicative of personality traits may not frequently appear in an author's writings. This highlights a critical nuance: While personality traits may indeed be reflected in an author's text, it does not necessarily mean that these markers are present in all texts authored by them—*the presence of lexical markers is not guaranteed to be ubiquitous*.

To reevaluate the idea supporting the relationship between personality and language use—APR, a comprehensive evaluation of all the elements involved in APR with all its complexities is necessary before it can be considered as a predictive or explanatory tool for personality traits. Consequently, in this work, we explore the limits of APR based on the two assumptions of *perceptibility* and *ubiquity*.

THE DATA-DRIVEN AUTOMATED PERSONALITY RECOGNITION PIPELINE

For most modern-day applications, the mere recognition of personality traits can enable computational devices to react differently to different people and situations. Therefore, the primary aim of text-based APR methods is to differentiate between the presence and absence of a personality trait within the text data using sophisticated AI methods. A typical AI system is comprised of data, labels and the learning method. Consequently, the scope of this section is to assemble a large quantity of text data with diverse data characteristics (topic, genre and domain), data acquisition protocol (controlled and uncontrolled), personality trait labels using different models (Big Five and Dark Triad) and evaluate the predictability of the labels from the provided text using a variety of AI methods. The methods were chosen to provide diversity in feature selection, popularity and representation. The diversity in these parameters would ensure that the learned information is generalizable and can enable the marginalized assessment of the validity of the APR assumptions. A typical NLP pipeline for APR is presented in Figure 1

20448295, 0, Downloaded from https://bpspsychub.onlinelibrary.wiley.com/doi/10.1111/bjop.12755 by University Of Arts London, Wiley Online Library on [21/01/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

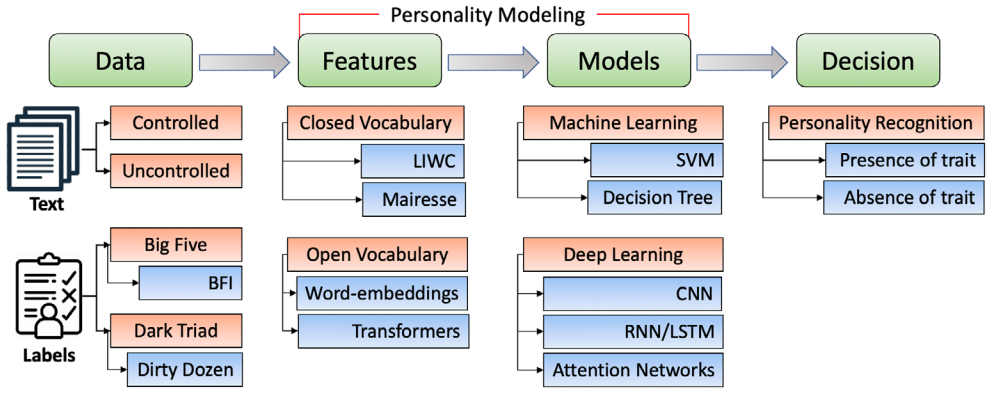


FIGURE 1 Natural language processing (NLP) pipeline for personality recognition.

With this section bridging the gap between personality psychology and AI, certain terminology and processes need to be clarified to maintain reader comprehension. In psychology, the personality trait of an individual is scored on a continuous 5-point Likert scale. On the 5-point Likert scale, a score below three indicates the absence of a trait and a score above three indicates the presence of the trait. An overall score of three indicates ambivalence or uncertainty. However, from a computational perspective, this problem is simplified further. A score above three is labelled trait-positive while lower scores are labelled trait-negative, resulting in a binary classification problem. Likewise, in psychology, the predictability of a trait is calculated through correlation measures, while the performance of AI systems is assessed using accuracy. In the general population, samples are not uniformly distributed across the presence and absence of the trait, especially while considering Dark Triad traits that are generally rarer. To account for this label imbalance while evaluating an APR system, a popular metric in AI—macro-average F1 score—is utilized. Macro-averaged F1 scores can assess the efficacy of AI systems without being subject to class imbalance. It can be interpreted similarly to accuracy.

Key components of the APR pipeline

Data

Data represent both the descriptive text data from an author and the associated personality trait scores for the author. In controlled settings, personality tests and questionnaires are administered with standardized instructions and conditions resulting in relatively good quality and structured text data. It was suggested that the more controlled and standardized the written sample, the easier it is to compare and learn personality trait relevant linguistic descriptors or features (Uher, 2013). Consequently, most datasets curated within the psychology community are designed with prompts specifically crafted to elicit emotions, expressions and opinions from participants. For instance, Essays—the gold-standard corpus with Big Five labels—required the participants to write a stream-of-consciousness (SOC) essay for 10 consecutive days and 20 min each day while the Sample14 corpus with Dark Triad labels required the participants to write text for three tasks: Write about who you are, SOC and Thematic Apperception Test (TAT) (Carey et al., 2015; Pennebaker & King, 1999). However, in this setting, the volume of data obtained is often limited and the data may not reflect how individuals behave and express themselves in real-world situations.

The real-world conditions, aptly called in-the-wild scenario, seldom offer fine-tuned control over data and present added challenges for robust APR evaluation. Text data collected in these conditions is typically noisy, less reliable and difficult to standardize. Despite these challenges, studies advocating for the use of such data suggest that it offers a genuine, uncensored glimpse into individuals' personality

20448295, 0, Downloaded from https://bpspsychub.onlinelibrary.wiley.com/doi/10.1111/bjop.12755 by University Of Arts London, Wiley Online Library on [21/01/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

TABLE 1 Description of datasets used for comparative assessment of APR methods.

Datasets (→)	WCPR13	WCPR14	Essays	Sample14	TwitterDT
Dataset code (→)	D1	D2	D3	D4	D5
Dataset statistics (↓)					
Personality model	Big Five	Big five	Big Five	Big Five & Dark Triad	Dark Triad
Label/Questionnaire type	IPIP-NEO	Observed Personality Labels	BFI	BFI & Dirty Dozen	Dirty Dozen
Source of data	Facebook Posts	Youtube vlog transcripts	Student Essays	Student Essays	Twitter posts
Total participants	243	404	2468	1126	652
Total text samples	7881	3205	2468	2252	544,729
Avg characters/sample	94	383	3296	3773	84
Avg words/sample	21	91	743	829	18
Avg sentences/sample	1	5	48	46	1

Note: D1—Celli et al., 2014, D2—Celli et al., 2013, D3—Pennebaker & King, 1999, D4—Carey et al., 2015, D5—Preotiuc-Pietro et al., 2016.
Abbreviations: BFI, Big Five Inventory; IPIP-NEO, International Personality Item Pool-Neuroticism, Extraversion and Openness.

traits through their daily expressions of thoughts, opinions and emotions (Carvalho & Pianowski, 2017; Celli et al., 2013, 2014). For instance, Facebook is an online platform meant not only for socializing but also for self-presentation (Lee et al., 2014). Assuming APR can be universally applied to all types of text, an effective AI system must overcome these obstacles to perform reliable personality recognition in-the-wild. To achieve this, include representative data from various modern sources, including social media posts, emails, blogs, and other unstructured formats, in our evaluation.

To compose a truly representative test bed, the selected corpora listed in Table 1 are considered to reflect personality traits either directly (via direct prompting) or indirectly (via expectation of self-presentation). Collectively, these represent diverse data characteristics (topic, genre and domain) and sample statistics (sample length, word/sentence count, etc), collected from various platforms/formats and under both strictly controlled and completely uncontrolled settings. Both Big Five and Dark Triad labels, extracted from popular self-reported questionnaires and expert annotations, are considered. Finally, to reduce observer bias and clutter, the long corpus names are substituted with short proxy labels (dataset code in Table 1) in the results and analysis. Consequently, selected corpora are representative of various scenarios faced in the real world and serve as a good benchmark set for an AI system focussed on APR.

Methods

Typically, discussions concerning methods in AI revolve around ML and DL techniques. An in-depth review of existing literature revealed that methods utilizing simple ML techniques such as Naive Bayes, Support Vector Machines (SVM) and Decision Trees to complex DL architecture have been utilized (Lima & de Castro, 2014; Mairesse et al., 2006, 2007; Oberlander & Nowson, 2006; Poria et al., 2013). DL methods that focus on spatial relations on words using Convolutional Neural Networks (CNN), sequential relationships using Recurrent Neural Networks (RNN), Long-short term memory (LSTM) and Gated Recurrent Unit (GRU), as well as directed relationships using Graph Neural Network (GNN) have been explored (Li et al., 2022; Majumder et al., 2017; Mehta et al., 2019; Sun et al., 2018; Wang et al., 2020). Attention neural networks have been explored to help the APR model focus on the parts of the text that carry the most informative personality-relevant information (Jiang et al., 2020; Xue et al., 2018; Zhou et al., 2022). Further more recent architecture such as transformers—the core architecture behind large language models such as ChatGPT—have

20448295, 0, Downloaded from https://bpspsychub.onlinelibrary.wiley.com/doi/10.1111/bjop.12755 by University Of Arts London, Wiley Online Library on [21/01/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

20448295, 0, Downloaded from https://bpspsychub.onlinelibrary.wiley.com/doi/10.1111/bjop.12755 by University Of Arts London, Wiley Online Library on [21/01/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

also been explored. However, in this study, the scope of transformers is limited to simple architectures such as BERT (Devlin et al., 2019). The most recent literature on advanced transformer models including ChatGPT are preliminary works that are yet to be peer-reviewed. However, for the curious reader, the APR performance reported for ChatGPT under simple prompting is comparable to the methods reported in this paper (Chen et al., 2023; Ji et al., 2023; Wen et al., 2023; Wu et al., 2023). With advances in prompting strategies, the performance of these SOTA LLMs will likely improve.

The perspective of greater importance for APR over ML and DL techniques is the features used to train these techniques. In simple terms, features can be considered a representation that highlights information that is critical to make a decision on the problem at hand. In our case, this critical information will be the trait-relevant parts of the text sample. Typically, features are either engineered or learned in AI systems.

Hand-engineered features infuse subject matter expertise and heuristics into the learning process. However, these methods typically require more time and effort for development, validation and optimization. The complementary approach to hand-engineered features relies on the automatic discovery of important features by learning relationships between the input data to the labels. A majority of the hand-engineered features utilize ML techniques for feature explainability and interpretability. Learned features are paired with DL techniques since both of them are black-box (not explainable nor directly interpretable).

Motivated by this nomenclature, the methods for APR are broadly categorized into two categories: Closed-vocabulary (CV) and Open-vocabulary (OV), representing feature-engineered and learned feature techniques respectively. CV approaches, also known as lexicon-based approaches, extract personality-relevant features by calculating the frequency distribution over various lexicon categories (Coltheart, 1981; Mohammad & Turney, 2010; Pennebaker et al., 2015). For instance, the most widely used lexicon dictionary in psychology and psycho-linguistic research communities—*LIWC*—categorizes words into explainable categories such as standard linguistic dimensions (e.g., punctuation, POS) and psychological word categories (e.g., affect, cognition, emotion) (Pennebaker et al., 2015). Similarly, the NRC Emotion Lexicon is widely used for its annotations of emotions and sentiments, offering insights into the affective aspects of personality traits (Mohammad & Turney, 2010). Despite being a knowledge-based approach, CV not only lacks flexibility as the relevant categories may become redundant from public discourse with time, but it is also difficult to adapt these lexicon dictionaries to other languages (Moskvichev et al., 2017). The limitation of this approach also includes the reduction in textual data to the mere frequency distribution of specific words while losing valuable syntactic and semantic information, which could prove crucial to identifying the sub-structures of text responsible for discerning personality traits. Furthermore, with a large number of words possible in any given language, exploring the utility of all words in the context of personality traits is a near-impossible task.

Based on the pretext that large amounts of data must contain sufficient personality characteristics, OV approaches choose the data mining route for APR by using LLM-derived word embeddings, topical preferences evaluated using Latent Dirichlet Allocation as well as various types of n-grams including words, characters as well as syntactical n-grams (Maharjan & Solorio, 2015; Schwartz et al., 2013; Stajner & Yenikent, 2020). Although relatively straightforward and hassle-free, this technique being completely data-driven may extract noisy features that may not provide any valuable information for APR. More recently, the hybrid approach combining knowledge-based lexicon categories along with data-driven features has been shown to significantly improve the performance of APR systems (Kazameini et al., 2020; Poria et al., 2013).

A diverse set of AI methods, described in Table 2, are selected for evaluation. This selection was guided by three main criteria: reproducibility, representativeness and recentness. The availability of software code, provided by the original authors of these AI methods, ensures reproducibility.[1] In terms of representativeness, OV methods constitute the majority of available AI techniques, followed by Hybrid and CV. We have strived to maintain this distribution in our selection to provide insights into the true state of APR. This is especially important for evaluating the recognition of Dark Triad

[1]Links to software code for AI methods are provided in Data S1.

TABLE 2 Brief description of the APR methods used for comparative assessment.

Type	Method	Description
CV	Boyd et al., 2022	LIWC-22 is a text analysis tool that calculates the frequency distribution over specialized word categories
	Mairesse et al., 2007	Built on LIWC-2001 and calculates utterance type, prosodic and MRC features
OV	Carducci et al., 2018	FastText embeddings classified with grid-search model optimization for SVM
	Sun et al., 2018	GloVe vectors trained on a combination of bi-directional LSTMs and CNN layers with 1,2,3-gram kernels
	Kampman et al., 2018	Word2Vec embeddings fed into CNN with 3,4,5-gram kernels
	Jiang et al., 2020	Sentence-level BERT transformer embeddings classified using CNN model with attention mechanism
	Mehta et al., 2020	First 512 words from text sample used to extract BERT transformer embeddings and classified using MLP classifier
	Jain et al., 2022	Deep neural model with BERT transformer finetuning followed by classification layer
	Y. Wang et al., 2023	Parameter-efficient fine-tuning with Low-Rank Adaptation (LoRA) of DeBERTa transformer model
Hybrid	Majumder et al., 2017	Document-level stylistic features (word 1,2,3-grams Word2Vec embeddings) and Mairesse features trained on a seven-layer CNN
	Kazameini et al., 2020	BERT transformer embeddings and Mairesse features extracted from chunks of 200 words classified using Bagged-SVM
	Yang et al., 2021	Tripartite graph attention learning framework using BERT and LIWC for contextual and psycholinguistic knowledge infusion.

personality traits due to its complicated nature and the dearth of methods in NLP literature specifically proposed for this personality model. Lastly, recent developments in AI have introduced advanced architectures, including graph-based models with attention mechanism and finetuned transformers. Our selection includes methods that embody these recent technological advancements.

Comprehensive evaluation of the APR pipeline

The comprehensive evaluation of the APR pipelines involves the benchmark of the chosen AI methods on the selected personality detection corpora. This evaluation is straightforward and serves three important purposes. *First*, the performance of APR is broken down at the method level revealing if additional heuristic/specialized knowledge is more useful than automatically discovering features by leveraging large-scale data sources. *Second*, the simultaneous assessment of two personality models—the Big Five and Dark Triad—comparisons can be drawn to evaluate the relative difficulty of applying APR to two diverging sets of traits. Existing literature in APR is focussed on recognizing the pro-social Big Five personality traits while its anti-social counterpart is often overlooked. *Finally*, the comprehensive evaluation can serve as a foundation for future APR methods to compare and baseline their efforts. Typically, the performance of APR approaches is merely reported from the original research works without testing their generalizability across corpora with varying textual characteristics and across different personality models (Artha Agastya et al., 2019; Feizi-Derakhshi et al., 2022; Mehta et al., 2019). Consequently, this comprehensive benchmark and its associated analysis serve a critical role in this work.

20448295, 0, Downloaded from https://bpspsychub.onlinelibrary.wiley.com/doi/10.1111/bjop.12755 by University Of Arts London, Wiley Online Library on [21/01/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Experiment configuration

The experiment protocol involves the application of the methods to the data. Initially pre-processing is performed to remove noisy/spurious information and standardize the corpus. Five corpora are assembled (D1, D2, D3 and D4 with Big Five and D4 and D5 with Dark Triad labels). During pre-processing, samples from languages other than English and containing less than five words each are removed. Emoticons, non-ASCII characters and URLs are also removed. Being a micro-blogging platform with sample sizes, TwitterDT (D5) was constrained to at least 100 tweets per author to ensure sufficient representative text data. The overall statistics of the pre-processed corpora are contained in Table 1.

In terms of personality labels, these corpora offered scores for every questionnaire item. The scores for items corresponding to each trait are averaged to get a single trait score. These trait scores are further subjected to thresholding at the mid-point (score 3 on the 5-point Likert scale) of the score range for getting the binary labels, where scores below the threshold represent negative class (absence of trait labelled with 0) while scores above the threshold represent positive class (presence of trait labelled with 1). For instance, the threshold for the BFI questionnaire is set to three as the scores range from one to five. Additionally, the Dark Triad personality model, a fourth cumulative dimension of the Dark Triad model, is computed by averaging the scores for independent traits.

Next, the pre-processed text data along with the thresholded labels are fed to the chosen method. This is considered a binary classification problem. As the CV methods only extract features from text data, multiple ML classifiers capable of binary classification are tested in search of an optimal classifier. The Decision Tree classifier is selected as the representative ML classifier for all CV methods since it performs most consistently on all corpora and traits. The Python library, scikit-learn[2], is used for the implementation of the Decision tree classifier without hyperparameter tuning and model optimization. Fivefold cross-validation is executed for all experiments to ensure stability and generalization of the reported results. The macro-averaged F1 score of the five cross-validation folds is reported.

Results and discussion

The point of notice in the benchmark analysis[3] is the mean performance and variation in performance on the detection of trait for each corpus. With a good selection of methods and diverse data, the experiment indicates the easiness and universality of extracting traits from the text. Three scenarios are likely to be observed.

1. Scenario 1: High average performance and low variance suggest that personality traits are easily detectable in text data.
2. Scenario 2: Low average performance[4] and low variance in performance indicate that personality traits are not evident in text data.
3. Scenario 3: Low average performance and high variance indicate the superiority of some APR methods over others.

Ideally, Scenario 1 is preferred with the expectation for a high performance involving accuracy or macro F1-scores surpassing the 0.90 value. For instance, in the case of SA, the performance reported for state-of-the-art methods normally exceeds macro-F1 scores of 0.9[5] with average lower-end performance exceeding a macro-F1 score of 0.7 (Wankhade et al., 2022). These results suggest

[2]https://scikit-learn.org

[3]Detailed performance assessment results for all corpora, traits, and APR approaches are provided in the Data S1.

[4]Low average performance is characterized by results that are close to the majority baseline.

[5]https://nlpprogress.com/english/sentiment–analysis.html

20448295, 0, Downloaded from https://bpspsychub.onlinelibrary.wiley.com/doi/10.1111/bjop.12755 by University Of Arts London, Wiley Online Library on [21/01/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

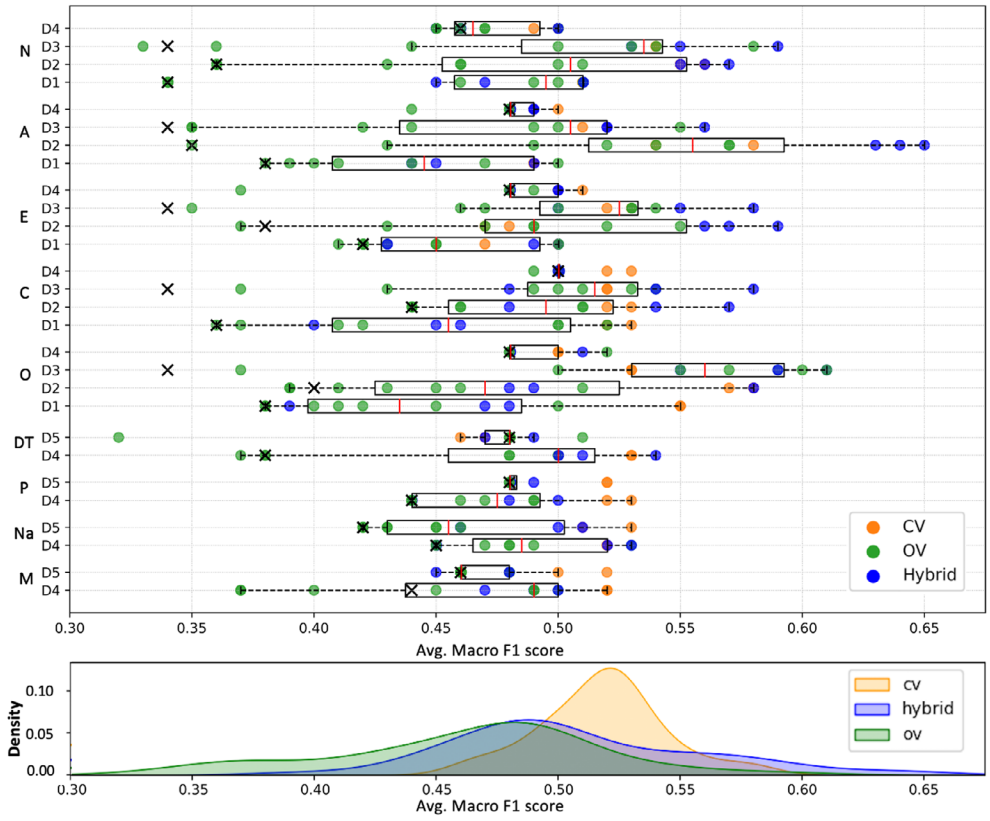


FIGURE 2 Figure illustrating the variance of performance using various personality detection methods where X marks Majority Baseline—assigning most frequent class labels to all examples. A, Agreeableness; C, Conscientiousness; DT, Cumulative Dark Triad; E, Extraversion; M, Machiavellianism; Na, Narcissism; N, Neuroticism; O, Openness; P, Psychopathy.

that sentiment is readily evident in the text. Consider a corpus with two Yelp-2013 reviews: 'The best donuts I've ever had' and 'I ordered the bbq wings and they were terrible' (Tang et al., 2015). In this example, the sentiment is clearly evident from the use of distinct positive and negative sentiment descriptors such as 'best' and 'terrible', respectively. Additionally, almost all occurrences of these descriptors in natural language data are likely to correspond with the same sentiment. If two new sentences are added to the corpus by replacing the adjectives 'finest' and 'appalling' in the first and second sentences, respectively, the model would possess more discriminating information for opposing sentiment polarities. Thus, these data augmentation would improve the model's generalizability and subsequently result in higher performance. In contrast, the performance of APR is bounded between 0.3 and 0.65 macro-F1 with many methods performing near the majority baseline. The results, depicted in Figure 2, highlight two important issues. First, the methods struggle to learn discriminating information about personality traits and often default to predicting the most frequent class label. Second, Scenarios 2 and 3 are observed with the traits not being evident in the text or indicate that some methods are more adept at detecting personality traits over others. This can be exemplified by the evaluation of trait Agreeableness on *D4* and *D2* representing Scenarios 2 and 3, respectively. *D4* exhibits median performance close to the majority baseline with low variance, whereas *D2* displays high variance in performance, with average results significantly deviating from the majority baseline. This suggestion is backed up by some more observations as mentioned below.

20448295, 0, Downloaded from https://bpspsychub.onlinelibrary.wiley.com/doi/10.1111/bjop.12755 by University Of Arts London, Wiley Online Library on [21/01/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

20448295, 0, Downloaded from https://bpspsychub.onlinelibrary.wiley.com/doi/10.1111/bjop.12755 by University Of Arts London, Wiley Online Library on [21/01/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

First, dark personality traits are harder to detect in text. Comparing the two personality models, higher variance is observed for the Big Five model than the Dark Triad personality model. This observation coupled with the markedly lower average performance for Dark Triad corpora implies that the dark psychological constructs of personality are more difficult to detect in text than the central Big Five traits.

Next, expert-encoded knowledge supports enhanced detection of personality traits in text. CV and Hybrid exhibit superior performance as compared to OV features (see the marginalized distribution at the bottom of Figure 2). This observation has also been noted in existing literature.

Finally, there is some degree of dependence of methods on the provided data characteristics, that is the methods do not work equally well on the same trait from a different corpus. For instance, the evaluation indicates high performance for Agreeableness on *D2*; Openness on *D1* and *D3*; cumulative Dark Triad on *D4* and Narcissism on *D5*. This could indicate that the lexical markers related to the trait are expressed strongly in those corpora.

Overall, the results indicate that AI-enabled methods are, to some extent, capable of detecting personality traits in text, though this task presents certain challenges. This highlights the need to critically reassess the underlying theoretical assumptions supporting APR.

VALIDATING THE LIMITS OF *APR*

The limits of the APR can be illustrated with the Dark Triad personality trait of Narcissism, which is often characterized by a desire to be the center of attention. It would be reasonable to assume that first-person singular pronouns (I-talk) would serve as a linguistic marker for this trait. Considering the three hypothetical statements 'I am a Mechanic', 'I am the best and most highly skilled Mechanic in town' and 'I am a Mechanic. I repair cars. I change and balance tires'. It is uncertain if the presence of 'I' in all sentences conveys the same meaning and, in turn, reflects Narcissism. Since the first sentence is merely factual, it could be indicative of the absence of narcissism. The second sentence could come across as arrogant and boastful and may indicate the presence of narcissistic tendencies. This variability demonstrates that personality may not be easily perceptible from the text. Moreover, the frequent occurrence of 'I' in the last sentence does not necessarily enhance the model's ability to identify personality traits accurately. This highlights a critical challenge: The ubiquity of certain linguistic markers does not guarantee precise personality assessments. Supporting this, research suggests that the presence of the first-person pronoun 'I' is not always associated with narcissism in natural language data, which is contrary to what might be expected (Carey et al., 2015).

With the experiments and discussions suggesting that the APR methods struggle to accurately identify personality traits from text, an in-depth quantitative assessment is required to establish the limits of APR. To this end, two lines of inquiry are necessary.

1. **Perceptibility**: Are the personality characteristics evident in text data?
2. **Ubiquity**: How often are the personality characteristics evident in the text data?

Perceptibility of personality in text

In AI terminology, a feature is said to be 'perceptible' if it is directly observable or measurable from the input data. Perceptible features are often preferred as they can provide a more straightforward and interpretable representation of the input data. For a feature to be perceptible in a binary classification task, it must have two main characteristics:

1. Discrminability: The feature should be able to discriminate between the two classes being analysed.
2. Consistency: The feature should be consistently associated with one of the classes.

In the previous example with Yelp-2013 reviews, the words 'best' and 'terrible' satisfy both criteria for perceptibility as they indicate positive and negative sentiments respectively, and are likely to be consistently associated with the same sentiment across natural language. On the other hand, the word 'I' in the narcissism example did not possess either of the characteristics required for perceptibility.

If the shallow text-based features, parts-of-speech, meet the two requirements for perceptibility, advanced AI methods can easily model the property of the data, eventually resulting in higher classification performance. Lexico-syntactic features such as POS tags are frequently used as shallow linguistic features to categorize words into their respective grammatical categories (Ebadat et al., 2011; Gamon, 2004; Gkotsis et al., 2014). Therefore, POS are chosen as the linguistic features to assess the perceptibility of personality in text data.

Discriminability

In computational linguistics, it is a common practice to evaluate the discriminability of classes by comparing the frequency of occurrence of the most common words between them. By itself, the difference in frequencies of words from opposite polarity classes does not provide information on the level of discriminability necessary for successful feature classification by a computational classifier. Hence, a benchmark representing optimal classification performance is necessary for comparison. As explained earlier in this section, since the SA methods have near-perfect performance, it is reasonable to assume that these data are highly discriminable. Therefore, the two SA corpora, Yelp-2013 and Yelp-2014, are utilized as a benchmark to evaluate the discriminability (Tang et al., 2015). In these corpora, a text sample is associated with a sentiment score evaluated on a 5-point Likert scale. To further accentuate the distinction between sentiment polarities for this study, a sentiment score of 1 represents the negative polarity class (label set to 0) while a sentiment score of 5 represents the positive polarity class (label set to 1). No pre-processing was performed on the texts from these corpora.

To determine the discriminability for a specific POS, the sum of absolute differences in relative frequencies of the most common words, also referred to as Area Under the Curve (AUC), can be computed. A comparison of the AUCs for sentiment and personality can then determine whether the discriminability of personality features is sufficient for successful classification. This is achieved in the following three steps. Step 1 involves the calculation of normalized frequencies of the most common words for the negative (neg) and positive (pos) polarity, respectively, followed by minimum sample sizes independently from sentiment and personality corpora as described in Algorithm 1. Recording the minimum sample size in Step 1 ensures that words with a minimum frequency are included at least once while conducting sampling distribution in the next step. Therefore, to ensure comparability, the sample size for sampling distribution is set as the maximum of the two sample sizes computed in Step 1. In Step 2, a sampling distribution is performed where the most common list of words for both sentiment and personality are sampled equally for a total of SampleSize times using the probabilities of word occurrences (normalized frequencies) calculated in Step 1. The newly sampled word list undergoes the same frequency calculation as in Step 1 followed by AUC calculation for both sentiment and personality as described in Algorithm 2.

Finally, in the last step, discriminability is assessed as the ratio of sentiment AUC to personality AUC, calculated using Equation 1, and denoted by δ. A δ greater than 1 would indicate a superior discriminability of sentiment over personality.

$$\delta = \frac{\text{SentimentAUC}}{\text{PersonalityAUC}} \tag{1}$$

20448295, 0, Downloaded from https://bpspsychub.onlinelibrary.wiley.com/doi/10.1111/bjop.12755 by University Of Arts London, Wiley Online Library on [21/01/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

ALGORITHM 1 Extract relative frequency and minimum sample size

1: **Input:** list of most common words W

2: neg ← relative frequency vector for W in the negative class.

3: pos ← relative frequency vector for W in the positive class.

4: $\text{min}_{\text{val}} \leftarrow \min(\text{neg}, \text{pos})$.

5: $\text{SampleSize}_{\text{min}} \leftarrow 10^{\lceil \log_{10}(\text{min}_{\text{val}}) \rceil}$.

6: **Return:** neg, pos, $\text{SampleSize}_{\text{min}}$.

ALGORITHM 2 Sampling Distribution

1: **Input:** list of most common words W; minimum sample sizes for sentiment (S_s) and personality (S_p); relative frequencies calculated in Step 1 pos and neg.

2: $\text{SampleSize} \leftarrow \max(S_s, S_p)$.

3: $W_{\text{pos}} \leftarrow \text{Sample}(W, \text{probability} = \text{pos}, \text{size} = \text{SampleSize})$.

4: $W_{\text{neg}} \leftarrow \text{Sample}(W, \text{probability} = \text{neg}, \text{size} = \text{SampleSize})$.

5: $\text{RelFreq}_{\text{pos}} \leftarrow [\text{freq}_{\text{word}} / \text{SampleSize for word} \in W_{\text{pos}}]$.

6: $\text{RelFreq}_{\text{neg}} \leftarrow [\text{freq}_{\text{word}} / \text{SampleSize for word} \in W_{\text{neg}}]$.

7: **Initialize** AUC ← 0.

8: **for** word **in** W **do**.

9: $\text{AUC} \leftarrow \text{AUC} + |\ \text{RelFreq}_{\text{pos}}[\text{word}] - \text{RelFreq}_{\text{neg}}[\text{word}]\ |$

10: **end for**.

Consistency

It refers to the association of discriminability with specific classes, lower consistency would suggest higher inherent variability in the data. Thus, for a corpus with randomly assigned labels, little to no discriminability and/or higher variability should be observed. To assess whether the discriminability (δ) is truly a property of the personality traits, the effect of variability in the language is computed by repeating the above experiment and replacing the sentiment corpus with the randomly shuffled labels of the same personality corpus. The ratio of AUC from randomly shuffled labels to the personality AUC is calculated using Equation 2, and denoted by ν. Since ν measures the variability in text, it can be interpreted as a confidence measure for δ. Therefore, lower ν would indicate higher confidence in the discriminability specifically due to personality labels. Ideally, for personality to be perceptible using a specific POS, δ should be less than 1 and ν closer to zero.

$$\nu = \frac{\text{RandomPersonalityAUC}}{\text{PersonalityAUC}} \tag{2}$$

20448295, 0, Downloaded from https://bpspsychub.onlinelibrary.wiley.com/doi/10.1111/bjop.12755 by University Of Arts London, Wiley Online Library on [21/01/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

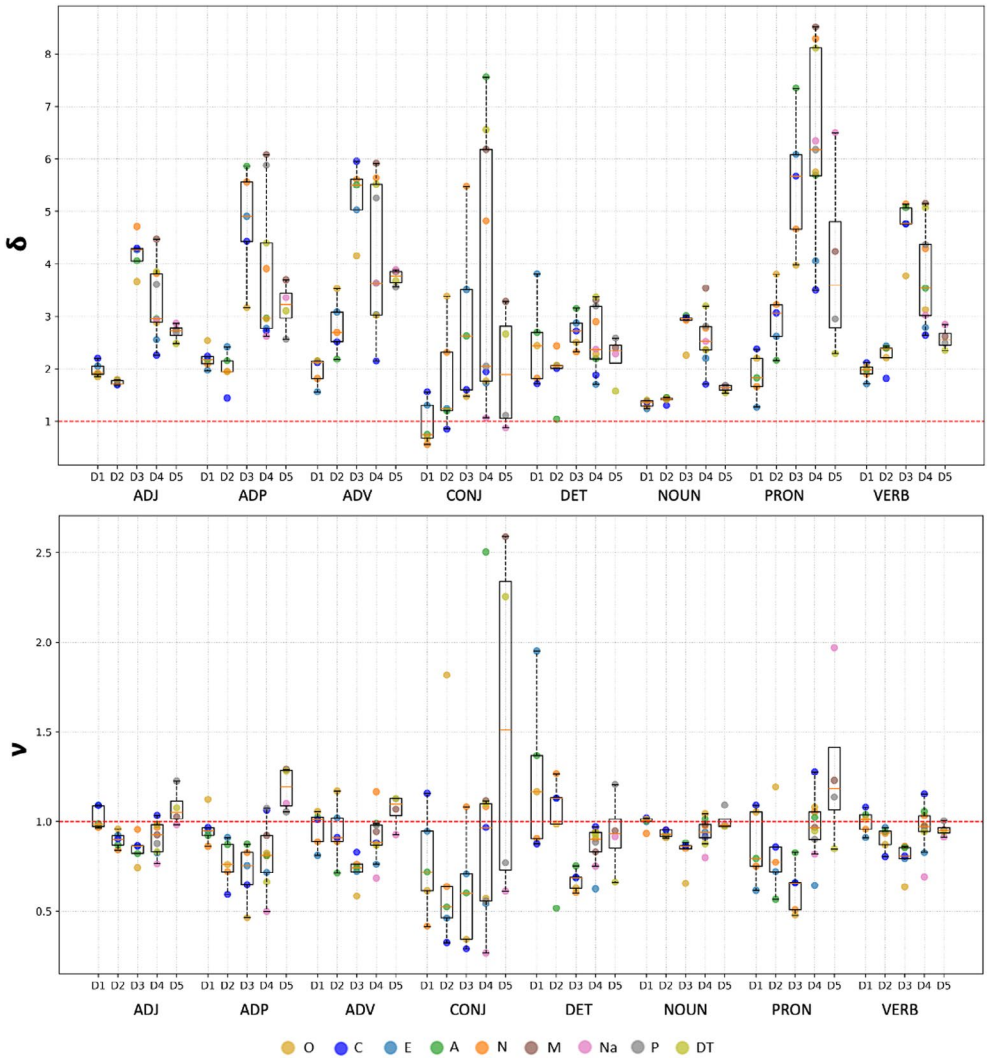


FIGURE 3 Discriminability (δ) and Variability (ν) in text data. A, Agreeableness; ADJ, Adjective; ADP, Adposition; ADV, Adverb; C, Conscientiousness; CONJ, Conjunction; DET, Determiner; DT, Cumulative Dark Triad; E, Extraversion; M, Machiavellianism; N, Neuroticism; Na, Narcissism; O, Openness; P, Psychopathy; PRON, Pronoun.

Experiment configuration

By employing the sampling distribution method, any potential effect of the differing number of most common words between sentiment and personality corpora on the comparison is eliminated. To further ensure comparability between sentiment and personality for the calculation of discriminability δ, an equal number of samples corresponding to personality classes are drawn from sentiment corpora. Finally, to prevent outliers caused by random sampling, the results are presented as an average of 20 runs with unique seeds. For the sake of generalizability, eight prominent POS tags are chosen for this analysis out of the universal POS tagset containing 12 POS tags (Petrov et al., 2011). The python library, NLTK[6] is used to extract POS tags from text.

[6]https://www.nltk.org/

20448295, 0, Downloaded from https://bpspsychub.onlinelibrary.wiley.com/doi/10.1111/bjop.12755 by University Of Arts London, Wiley Online Library on [21/01/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Results and discussion

From the results[7] depicted in Figure 3, two important observations can be made. First, there is a consistently superior discriminability of sentiment over personality across most cases, as indicated by δ values consistently above 1, except in some instances. This difference is reflected in the significant disparities in state-of-the-art (SOTA) performance between the two fields. Additionally, the parts of speech (POS) contributing to discriminability vary across different corpora. For example, in dataset *D1*, conjunctions (*CONJ*) are most discriminable for Neuroticism and least for Conscientiousness. Conversely, in dataset *D2*, *CONJ* is most discriminable for Conscientiousness and least for Agreeableness, highlighting how the availability of linguistic features significantly affects discriminability between datasets.

Secondly, except in some cases such as CONJ in *D5* and DET in *D1*, ν equal to or less than 1 suggests that, albeit weak, POS possesses information that could potentially enhance the perceptibility of personality. For instance, consider CONJ in *D1* where the δ is less than 1 and ν is less than 0.5 suggesting the usability of CONJ for discriminating between traits such as Neuroticism and Agreeableness in this corpus. However, the variance in observations across corpora, traits, and POS discourages the 'one-shoe-fits-all' strategy that lexicon-based APR methods use.

In general, *the results of this experiment suggest that the first assumption of APR, which states that personality characteristics are perceptible in text, is not always valid.*

Ubiquity of personality in language

Personality can be considered 'ubiquitous' in text if the presence of personality characteristics increases proportionally with the volume of text data. The presence of personality characteristics in the text can be assessed using a collective decision of APR methods employing variable personality recognition techniques while the volume of data can be measured using the number of words and sentences in a text sample. As personality traits are not easily evident in text data, a higher volume of such data is not expected to be strongly associated with the performance of APR methods. However, an empirical evaluation is necessary to confirm this conjecture.

Two values are computed for each text sample: the probability of accurate classification and the amount of data represented by the number of words and sentences. For each algorithm, let C_i be a binary label that is 1 if the sample is classified correctly by that algorithm, and 0 otherwise, and n be the number of algorithms used to classify the sample. Then, the probability of accurate classification for the sample, $\mathrm{P_{acc}}$, is given by Equation 3

$$\mathrm{P_{acc}} = \frac{1}{n}\sum_{i=1}^{n} C_i \tag{3}$$

The algorithms that analysed the original document in chunks, specifically Jiang20 and Kazameini20, were excluded from this analysis. Finally, Pearson's correlation analysis was conducted to assess the relationship between the volume of data (measured by the number of words and sentences) and the probability of accurate classification. The results of this analysis are presented in Table 3. If there is a significantly strong correlation between the volume of data and the probability of accurate classification, it would suggest that as the amount of data increases, the likelihood of personality characteristics being present also increases.

[7]Extended version of the results for perceptibility analysis can be found in the Data S1.

20448295, 0, Downloaded from https://bpspsychub.onlinelibrary.wiley.com/doi/10.1111/bjop.12755 by University Of Arts London, Wiley Online Library on [21/01/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

TABLE 3 Pearson's correlation between Volume of data and likelihood of correct personality classification.

Dataset	Counts (↓) trait (→)	O	C	E	A	N
D1	Words	0.02	−0.09	−0.01	−0.03	**−0.20****
	Sentences	0.01	−0.07	0.03	0.00	**−0.20****
D2	Words	0.05	0.05	0.02	**0.13***	0.01
	Sentences	0.06	−0.04	−0.03	**0.13****	0.03
D3	Words	0.03	−0.03	0.02	0.00	−0.01
	Sentences	**0.04***	−0.02	−0.01	0.01	−0.02
D4	Words	−0.01	−0.03	−0.03	−0.01	0.02
	Sentences	0.00	0.00	−0.01	−0.01	0.00
	Trait (→)	**M**	**Na**		**P**	**DT**
D4	Words	0.01	**0.06****		0.04	−0.01
	Sentences	0.00	**0.06****		0.03	−0.01
D5	Words	−0.03	−0.05		0.03	0.01
	Sentences	−0.04	−0.02		0.04	0.03

Note: Statistical significance of Pearson's correlation $p < 0.05$ is denoted by *, and $p < 0.01$ by **. Bold values represent statistically significant results.

Abbreviations: A, Agreeableness; C, Conscientiousness; DT, Cumulative Dark Triad; E, Extraversion; M, Machiavellianism; Na, Narcissism; N, Neuroticism; O, Openness; P, Psychopathy.

Results

The results indicate that for most personality traits, there is no substantial correlation between the volume of data and the probability of accurate classification, except for a few traits. Among the significantly correlated traits, most have a weak, almost non-existent correlation. While a weak but statistically significant correlation is observed between Agreeableness in *D2* and Neuroticism in *D1*, the contradictory trends cast doubt on the effectiveness of using a larger volume of data to enhance personality characteristics in text.

Overall, this *experiment contradicts the second assumption of APR which suggests that having a substantial amount of natural language data increases the likelihood of accurately capturing personality traits.*

BEYOND THE LEXICAL LIMIT: OPEN PROBLEMS AND OPPORTUNITIES

The tools and methods that are developed in AI can be used for analysis, visualization and even explaining results from data-driven methods. These are very useful and powerful techniques. However, AI still operates under the assumption that the 'data' provided to the learning system and the learning "method" chosen by the expert fits the problem. The application of APR introduced in this paper to support behavioural psychology is one such area where the expectations of the AI system are not met, and its detrimental effects are visible. Several 'methods' have been tried and tested ranging from simple lexicon-based classification model to LLMs like ChatGPT but their efficacy is severely limited if the 'data' does not represent personality traits sufficiently.

Our experimental findings offer practical suggestions for enhancing APR by addressing limitations related to the "data".

Given the typically limited availability of text samples for personality assessment—often just a single sample per author—future efforts should prioritize collecting more extensive data. Incorporating multi-text analysis could significantly enhance the robustness and reliability of APR methods.

20448295, 0, Downloaded from https://bpspsychub.onlinelibrary.wiley.com/doi/10.1111/bjop.12755 by University Of Arts London, Wiley Online Library on [21/01/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Additionally, the observation that personality traits are not readily perceptible in text highlights the need for careful selection of data sources, particularly for 'in-the-wild' scenarios. Platforms where users frequently engage in in-depth debates and discussions are more likely to yield personality-relevant information (Ryan & Xenos, 2011; Vaid & Harari, 2021). Such platforms can facilitate active and emotionally expressive exchanges that are rich in personality indicators, unlike Twitter, where our results indicate weaker performance. For platforms such as Twitter, where content is typically less expressive, focussing on emotionally charged or highly relevant topics—such as public health crises or geopolitical events—can enhance the likelihood of capturing expressive content that reflects personality traits more accurately.

Moreover, our results suggest that text alone may not always provide a sufficient basis for assessing personality. Incorporating additional data types such as video (Escalante et al., 2020; Giritlioğlu et al., 2021; Sun et al., 2022), along with metadata such as age and gender (Preoţiuc-Pietro et al., 2015; Schwartz et al., 2013), and employing methods such as Social Network Analysis (Souri et al., 2018; Sun et al., 2023), can enrich the context and enhance the discriminative capability of AI models, leading to more reliable APR.

Additionally, our observations indicate that personality traits are not ubiquitously reflected in text. Leveraging the Natural Language Inference (NLI) capabilities of LLMs such as ChatGPT to selectively extract the most personality-relevant portions of text can significantly reduce personality-irrelevant content or noise from text. This targeted extraction approach may ensure that the analysed text more accurately mirrors the underlying personality traits, thereby improving the effectiveness of APR in practical applications.

While the AI community can integrate the above suggestions into APR systems, the psychology community can also play a crucial role by refining the data with insights from subject matter experts—psychologists, to a point where it can be easily and effectively handled by the AI system. To this end, three open problems have been identified. Each problem refines the data from different perspectives using knowledge provided by psychology research. These open problems are discussed below.

Knowledge encoding

The benchmarks and analysis performed in this article suggest that CV and hybrid techniques are more effective at performing APR than OV methods. The CV technique leverages specialized lexicons, such as the LIWC, that incorporate lexicon categories based on their respective psycho-linguistic properties in text. However, these lexical repositories are manually curated with a significant amount of effort. AI techniques can enable the identification of words that correlate with these specialized linguistic descriptors. For instance, words used in similar contexts such as those identified by BERT. Similarly, some recent works have shown promise in utilizing specialized graph representations curated by psychologists that aim to encode subject matter expertise into the APR pipeline (Johnson & Murty, 2023; Liu et al., 2024; Ramezani et al., 2022; Yan et al., 2024; Zhu et al., 2022). However, the design and development of these knowledge representations are weakly reasoned and supported. A more general, well-constrained, and growing knowledge repository, like that achieved by SenticNet (Cambria et al., 2010), will be key for effective APR. A synergy between psychology and AI can help accelerate the process of encoding and processing information effortlessly while also reliably and accurately recognizing personality from text.

Explainable feature development

AI systems are very adept at identifying discriminating features. But these may be capturing an artefact in text that does not necessarily have any relationship to personality traits. However, the inclusion of these spurious features in the learning process can adversely affect the performance of the system-ultimately affecting the generalizability of the models. For instance, the Big Five personality

20448295, 0, Downloaded from https://bpspsychub.onlinelibrary.wiley.com/doi/10.1111/bjop.12755 by University Of Arts London, Wiley Online Library on [21/01/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

model represents five independent dimensions of personality. However, several APR methods assume co-dependence between these traits suggesting that although the traits themselves may be independent, their representation in text might be correlated. Currently, there are no guidelines or heuristics grounded in psychology that confirm this phenomenon. A human-AI teaming—where the AI systems discover the features that are effective at detecting personality traits and human experts that can refine this learned knowledge is key for the development of effective APR approaches. Explainable AI (XAI) techniques can help break down learned features, emphasize their influence on personality trait predictions, and identify spurious correlations (Panfilova & Turdakov, 2024). Psychologists and subject matter experts can leverage the insights provided by XAI—through visualizations and detailed feature explanations—to guide model refinement. This collaborative approach ensures that models align with psychological principles and avoid dependence on misleading patterns.

Scope expansion beyond personality

Implementations of AI systems for APR typically consider personality a stand-alone problem. However, existing research in psychology points towards the role emotions may play in personality recognition. It was found that the removal of emotionally neutral sentences greatly enhances the method's performance in detecting Big Five traits (Majumder et al., 2017). Furthermore, other methods incorporated affective states such as emotion and sentiment to enhance the performance of the APR methods (Darliansyah et al., 2019; Mohammad & Kiritchenko, 2013; Ren et al., 2021; Yan et al., 2024). Similarly, there may be other areas of psychology related to personality that may play a role in clarifying latent trait-relevant linguistic descriptors to the AI system.

In summary, *to step beyond the limitation of a purely data-driven methodology, knowledge infusion* via *human-AI teaming is essential*. The suggestions introduced above are only a few exemplary ways to achieve this goal.

LIMITATIONS

Beyond the aspects covered in this study—perceptibility and ubiquity of personality traits in text as well as personality-informativeness of text data, it is likely that modelling choices and feature representations equally affect the automatic recognition of personality traits in text.

First, one could argue that further exploration of better feature representations might improve model performance. However, we have tested features that reflect the current state of NLP and APR, including syntactic, lexical, contextual and knowledge-guided features. Thus, our findings encompassing a broad and representative range of feature types available in NLP and used in APR studies are unlikely to be affected by additional feature representations. For meaningful performance improvements beyond this, more advanced manual feature engineering or human–AI collaboration would likely be required, as suggested in Section 5.2.

Additionally, it is plausible that models performing well on specific datasets and traits are, in part, learning dataset-specific characteristics rather than generalizable patterns. This issue has been highlighted in model explainability research, where misleading SOTA results were obtained by exploiting dataset-specific characteristics (Aminimehr et al., 2024). Similarly, in Scenario 3 as outlined in Section 3.2.2, we considered that a combination of low average performance and high variance might suggest the superiority of certain methods over others. However, these results could potentially be influenced by dataset-specific spurious correlations. Conversely, it is well-documented that certain dataset-specific characteristics can naturally amplify the expression of specific traits. For instance, text samples from Facebook often exhibit more pronounced personality markers for traits extroversion and openness (Farnadi et al., 2013; Park et al., 2015). This indicates that while variations in model performance across datasets may, in some cases, be influenced by spurious correlations, they are not entirely so. However, the diversity and scale of APR methods tested in this study help to reduce the likelihood of these issues

20448295, 0, Downloaded from https://bpspsychub.onlinelibrary.wiley.com/doi/10.1111/bjop.12755 by University Of Arts London, Wiley Online Library on [21/01/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

affecting the overall conclusions. Conducting a thorough error analysis can further help clarify and address these concerns.

Although these observations do not change our conclusion regarding the current limitations of APR methods in reliably recognizing personality traits, it nevertheless underscores the need for detailed feature and error analysis to better understand the relevance and validity of features learned by the models for accurate personality predictions—a limitation of our study. Such an investigation, well-suited for XAI techniques, was beyond the scope of this study but represents a critical direction for future research.

CONCLUSION

Data-driven methods have consistently surpassed traditional techniques in several research fields. This trend was fueled by powerful AI methods getting trained on large amounts of diverse data openly available on online platforms using affordable and accessible large-scale computational resources. Despite having access, the trend did not hold for APR. An in-depth analysis from the perspectives of different AI methods and corpora, collected from existing literature, revealed some interesting observations. First, a significant increase in recognition performance was obtained when the methods were provided access to expert encoded knowledge representation. Next, variation in recognition performance was observed for Big Five and Dark Triad traits extracted from the same corpus. Additionally, variation in recognition performance was also observed for methods operating on the same personality trait in different corpora. These observations suggested that the provided text data were not sufficiently representing the linguistic markers required to detect personality in text accurately in contrast to the expectations of personality assessment from language use—APR.

Summarizing the assumptions of APR into key concepts of 'perceptibility' and 'ubiquity' representing the 'ease of interpretation' and 'ease of acquisition' from text respectively, several experiments were designed to evaluate whether these key concepts were always satisfied in text and, in turn, evaluate the validity of personality assessment from text. It was found that the assumptions underlying the relationship between personality and language expression are not universally valid. Consequently, several suggestions were put forth for the AI and Psychology communities as open problems to bypass the performance limit by targeted data collection, leveraging advanced AI techniques, encoding more expert knowledge into the problem, and introducing human–AI teaming to resolve spurious knowledge and features to eventually leverage the advantage of AI for APR.

AUTHOR CONTRIBUTIONS

Avanti Bhandarkar: Conceptualization; data curation; software; methodology; formal analysis; visualization; writing – original draft; writing – review and editing. **Ronald Wilson:** Conceptualization; investigation; supervision; writing – review and editing. **Anushka Swarup:** Software; validation; visualization. **Gregory D. Webster:** Project administration; supervision; writing – review and editing. **Damon Woodard:** Project administration; supervision.

ACKNOWLEDGEMENTS

We would like to extend our gratitude to Dr. James Pennebaker and Dr. Mattias Mehl for providing the Essays and Sample14 datasets. We would also like to thank Dr. Daniel Preotiuc-Pietro for sharing the TwitterDT dataset.

CONFLICT OF INTEREST STATEMENT

The authors have no conflicts of interest to declare.

DATA AVAILABILITY STATEMENT

Data sharing is not applicable to this article as no new data were created or analyzed in this study.

20448295, 0, Downloaded from https://bpspsychub.onlinelibrary.wiley.com/doi/10.1111/bjop.12755 by University Of Arts London, Wiley Online Library on [21/01/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

ORCID

Avanti Bhandarkar https://orcid.org/0000-0002-9297-5029

REFERENCES

Allport, G. W. (1937). *Personality: A psychological interpretation*. Holt.

Aminimehr, A., Khani, P., Molaei, A., Kazemeini, A., & Cambria, E. (2024). *Thexplain: A text-based explanation method for scene classification models with the statistical prediction correction*. Proceedings of the Conference on Governance, Understanding and Integration of Data for Effective and Responsible AI (pp. 54–60). Association for Computing Machinery. https://doi.org/10.1145/3665601.3669841

Artha Agastya, I. M., Dwi Handayani, D. O., & Mantoro, T. (2019). *A systematic literature review of deep learning algorithms for personality trait recognition*. Proceeding 5th international conference on computing engineering and design (ICCED) (pp. 1–16). IEEE. https://doi.org/10.1109/icced46541.2019.9161107

Boyd, R. L., Ashokkumar, A., Seraj, S., & Pennebaker, J. W. (2022). *The development and psychometric properties of liwc-22* (tech. rep.). University of Texas at Austin.

Cambria, E., Speer, R., Havasi, C., & Hussain, A. (2010). Senticnet: A publicly available semantic resource for opinion mining. *2010 AAAI Fall Symposium Series*, 14–18. Technical Report FS-10-02.

Carducci, G., Rizzo, G., Monti, D., Palumbo, E., & Morisio, M. (2018). TwitPersonality: Computing personality traits from tweets using word embeddings and supervised learning. *Information*, *9*(5), 127. https://doi.org/10.3390/info9050127

Carey, A. L., Brucks, M. S., Küfner, A. C. P., Holtzman, N. S., Große Deters, F., Back, M. D., Donnellan, M. B., Pennebaker, J. W., & Mehl, M. R. (2015). Narcissism and the use of personal pronouns revisited. *Journal of Personality and Social Psychology*, *109*(3), e1–e15. https://doi.org/10.1037/pspp0000029

Carvalho, L. D. F., & Pianowski, G. (2017). Pathological personality traits assessment using facebook: Systematic review and meta-analyses. *Computers in Human Behavior*, *71*, 307–317. https://doi.org/10.1016/j.chb.2017.01.061

Cattell, R. B. (1950). *Personality: A systematic theoretical and factual study* (1st ed.). McGraw-Hill. https://doi.org/10.1037/10773-000

Celli, F., Lepri, B., Biel, J.-I., Gatica-Perez, D., Riccardi, G., & Pianesi, F. (2014). *The workshop on computational personality recognition 2014*. Proceeding 22nd ACM international conference on multimedia (pp. 1245–1246). Association for Computing Machinery. https://doi.org/10.1145/2647868.2647870

Celli, F., Pianesi, F., Stillwell, D., & Kosinski, M. (2013). *Workshop on computational personality recognition: Shared task*. Proceeding International AAAI Conference on Web and Social Media (Vol. *7*, pp. 2–5). AAAI Press. https://doi.org/10.1609/icwsm.v7i2.14467

Chen, L., Wu, Y., Li, Q., Song, Y., Yuan, C., & Zeng, D. (2023). Mining the user's personality with an attention-based label prompt method. *IEEE Intelligent Systems*, *1–8*, 31–39. https://doi.org/10.1109/MIS.2023.3320445

Coltheart, M. (1981). The mrc psycholinguistic database. *The Quarterly Journal of Experimental Psychology Section A*, *33*(4), 497–505. https://doi.org/10.1080/14640748108400805

Darliansyah, A., Naeem, M. A., Mirza, F., & Pears, R. (2019). Sentipede: A smart system for sentiment-based personality detection from short texts. *Journal of Universal Computer Science*, *25*(10), 1323–1352.

Devlin, J., Chang, M. W., Lee, K., & Toutanova, K. (2019, June). BERT: Pre-training of deep bidirectional transformers for language understanding. In *Proceedings of the 2019 conference of the North American chapter of the association for computational linguistics: Human language technologies, Vol. 1 (Long and Short Papers)* (pp. 4171–4186).

Ebadat, A. R., Claveau, V., & Sébillot, P. (2011). Using shallow linguistic features for relation extraction in bio-medical texts. In *Traitement Automatique des Langues Naturelles, TALN* (pp. 125–130).

Escalante, H. J., Kaya, H., Salah, A. A., Escalera, S., Güçlütürk, Y., Güçlü, U., Baró, X., Guyon, I., Junior, J. C. J., Madadi, M., & Ayache, S. (2020). Modeling, recognizing, and explaining apparent personality from videos. *IEEE Transactions on Affective Computing*, *13*(2), 894–911.

Eysenck, H. (2013). *The structure of human personality (psychology revivals)*. Routledge. https://doi.org/10.4324/9780203753439

Farnadi, G., Zoghbi, S., Moens, M.-F., & de Cock, M. (2013). *Recognising personality traits using facebook status updates*. Proceedings of the International AAAI Conference on Web and Social Media (Vol. *7*, pp. 14–18). AAAI Press.

Feizi-Derakhshi, A.-R., Feizi-Derakhshi, M.-R., Ramezani, M., Nikzad-Khasmakhi, N., Asgari-Chenaghlu, M., Akan, T., Ranjbar-Khadivi, M., Zafarni-Moattar, E., & Jahanbakhsh-Naghadeh, Z. (2022). Text-based automatic personality prediction: A bibliographic review. *Journal of Computational Social Science*, *5*(2), 1555–1593. https://doi.org/10.1007/s42001-022-00178-4

Gamon, M. (2004). *Linguistic correlates of style: Authorship classification with deep linguistic analysis features*. Proceedings 20th international conference on computational linguistics, COLING 2004 (pp. 611–617). Association of Computational Linguistics. https://aclanthology.org/C04-1088

Giritlioğlu, D., Mandira, B., Yilmaz, S. F., Ertenli, C. U., Akgür, B. F., Kınıklıoğlu, M., Kurt, A. G., Mutlu, E., Gürel, Ş. C., & Dibeklioğlu, H. (2021). Multimodal analysis of personality traits on videos of self-presentation and induced behavior. *Journal on Multimodal User Interfaces*, *15*(4), 337–358.

Gkotsis, G., Stepanyan, K., Pedrinaci, C., Domingue, J., & Liakata, M. (2014, June). It's all in the content: State of the art best answer prediction based on discretisation of shallow linguistic features. In *Proceedings of the 2014 ACM conference on web science* (pp. 202–210).

20448295, 0, Downloaded from https://bpspsychub.onlinelibrary.wiley.com/doi/10.1111/bjop.12755 by University Of Arts London, Wiley Online Library on [21/01/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Goldberg, L. R. (1981). Language and individual differences: The search for universals in personality lexicons. In L. Wheeler (Ed.), *Review of personality and social psychology* (Vol. *2*, pp. 141–165). Sage Publication.

Goldberg, L. R. (1993). The structure of phenotypic personality traits. *American Psychologist*, *48*(1), 26–34. https://doi.org/10.1037/0003-066x.48.1.26

Goldberg, L. R. (1999). A broad-bandwidth, public domain personality inventory measuring the lower-level facets of several five-factor models. In I. Mervielde, I. Deary, F. De Fruyt, & F. Ostendorf (Eds.), *Personality psychology in Europe* (Vol. *7*, pp. 7–28). Tilburg University Press.

Jain, D., Kumar, A., & Beniwal, R. (2022). Personality bert: A transformer-based model for personality detection from textual data. In A. K. Bashir, G. Fortino, A. Khanna, & D. Gupta (Eds.), *Lecture notes in networks and systems* (pp. 515–522). Springer Nature Singapore. https://doi.org/10.1007/978-981-19-0604-6_48

Ji, Y., Wu, W., Zheng, H., Hu, Y., Chen, X., & He, L. (2023). *Is chatgpt a good personality recognizer? A preliminary study*. Cornell University. https://doi.org/10.48550/arXiv.2307.03952

Jiang, H., Zhang, X., & Choi, J. D. (2020). Automatic text-based personality recognition on monologues and multiparty dialogues using attentive networks and contextual embeddings (student abstract). *Proceedings AAAI Conference on Artificial Intelligence*, *34*(10), 13821–13822. https://doi.org/10.1609/aaai.v34i10.7182

John, O. P., Angleitner, A., & Ostendorf, F. (1988). The lexical approach to personality: A historical review of trait taxonomic research. *European Journal of Personality*, *2*(3), 171–203. https://doi.org/10.1002/per.2410020302

John, O. P., Donahue, E. M., & Kentle, R. L. (1991). Big Five Inventory. *Journal of Personality and Social Psychology*, *1*, 20. https://doi.org/10.1037/t07550-000

Johnson, S. J., & Murty, M. R. (2023). An aspect-aware enhanced psycholinguistic knowledge graph-based personality detection using deep learning. *SN Computer Science*, *4*(3), 293.

Jonason, P. K., & Webster, G. D. (2010). The dirty dozen: A concise measure of the dark triad. *Psychological Assessment*, *22*(2), 420–432. https://doi.org/10.1037/a0019265

Kampman, O., Barezi, E. J., Bertero, D., & Fung, P. (2018). Investigating audio, video, and text fusion methods for end-to-end automatic personality prediction. In *Proceedings of the 56th annual meeting of the association for computational linguistics (Vol. 2: Short papers)* (pp. 606–611).

Kazameini, A., Fatehi, S., Mehta, Y., Eetemadi, S., & Cambria, E. (2020). Personality trait detection using bagged SVM over BERT word embedding ensembles. *arXiv preprint arXiv:2010.01309*.

Koutsoumpis, A., Oostrom, J. K., Holtrop, D., van Breda, W., Ghassemi, S., & de Vries, R. E. (2022). The kernel of truth in text-based personality assessment: A meta-analysis of the relations between the big five and the linguistic inquiry and word count (liwc). *Psychological Bulletin*, *148*(11–12), 843–868. https://doi.org/10.1037/bul0000381

Lee, E., Ahn, J., & Kim, Y. J. (2014). Personality traits and self-presentation at facebook. *Personality and Individual Differences*, *69*, 162–167. https://doi.org/10.1016/j.paid.2014.05.020

Li, Y., Kazemeini, A., Mehta, Y., & Cambria, E. (2022). Multitask learning for emotion and personality traits detection. *Neurocomputing*, *493*, 340–350. https://doi.org/10.1016/j.neucom.2022.04.049

Lima, A. C. E., & de Castro, L. N. (2014). A multi-label, semi-supervised classification approach applied to personality prediction in social media. *Neural Networks*, *58*, 122–130. https://doi.org/10.1016/j.neunet.2014.05.020

Liu, W., Sun, Z., Wei, S., Zhang, S., Zhu, G., & Chen, L. (2024). Ps-gcn: Psycholinguistic graph and sentiment semantic fused graph convolutional networks for personality detection. *Connection Science*, *36*(1), 2295820.

Maharjan, S., & Solorio, T. (2015). Using wide range of features for author profiling. *Conference and Labs of the Evaluation Forum (CLEF)*, *1391*.

Mairesse, F., & Walker, M. (2006). Words mark the nerds: Computational models of personality recognition through language. In *Proceedings of the annual meeting of the cognitive science society* (Vol. 28, No. 28).

Mairesse, F., Walker, M. A., Mehl, M. R., & Moore, R. K. (2007). Using linguistic cues for the automatic recognition of personality in conversation and text. *Journal of Artificial Intelligence Research*, *30*, 457–500. https://doi.org/10.1613/jair.2349

Majumder, N., Poria, S., Gelbukh, A., & Cambria, E. (2017). Deep learning-based document modeling for personality detection from text. *IEEE Intelligent Systems*, *32*(2), 74–79. https://doi.org/10.1109/MIS.2017.23

Mehta, Y., Fatehi, S., Kazameini, A., Stachl, C., Cambria, E., & Eetemadi, S. (2020). Bottom-up and top-down: Predicting personality with psycholinguistic and language model features. *IEEE International Conference on Data Mining (ICDM)*, *2020*, 1184–1189. https://doi.org/10.1109/ICDM50108.2020.00146

Mehta, Y., Majumder, N., Gelbukh, A., & Cambria, E. (2019). Recent trends in deep learning based personality detection. *Artificial Intelligence Review*, *53*(4), 2313–2339. https://doi.org/10.1007/s10462-019-09770-z

Mohammad, S., & Kiritchenko, S. (2013). *Using nuances of emotion to identify personality*. Proceedings of the International AAAI Conference on Web and Social Media (Vol. *7*, pp. 27–30). AAAI Press.

Mohammad, S., & Turney, P. (2010). *Emotions evoked by common words and phrases: Using mechanical turk to create an emotion lexicon*. Proceedings 2010 Workshop on Computational Approaches to Analysis and Generation of Emotion in Text (pp. 26–34). Association of Computational Linguistics. https://aclanthology.org/W10-0204

Moreno, J. D., Martínez-Huertas, J. Á., Olmos, R., Jorge-Botana, G., & Botella, J. (2021). Can personality traits be measured analyzing written language? A meta-analytic study on computational methods. *Personality and Individual Differences*, *177*, 110818. https://doi.org/10.1016/j.paid.2021.110818

20448295, 0, Downloaded from https://bpspsychub.onlinelibrary.wiley.com/doi/10.1111/bjop.12755 by University Of Arts London, Wiley Online Library on [21/01/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Moskvichev, A., Dubova, M., Menshov, S., & Filchenkov, A. (2017). Using linguistic activity in social networks to predict and interpret dark psychological traits. In *Artificial intelligence and natural language* (pp. 16–26). Springer International Publishing. https://doi.org/10.1007/978-3-319-71746-3_2

Oberlander, J., & Nowson, S. (2006). Whose thumb is it anyway? Classifying author personality from weblog text. In *Proceedings of the COLING/ACL 2006 main conference poster sessions* (pp. 627–634).

Panfilova, A. S., & Turdakov, D. Y. (2024). Applying explainable artificial intelligence methods to models for diagnosing personal traits and cognitive abilities by social network data. *Scientific Reports*, *14*, 5369. https://doi.org/10.1038/s41598-024-56080-8

Park, G., Schwartz, H. A., Eichstaedt, J. C., Kern, M. L., Kosinski, M., Stillwell, D. J., Ungar, L. H., & Seligman, M. E. P. (2015). Automatic personality assessment through social media language. *Journal of Personality and Social Psychology*, *108*(6), 934–952. https://doi.org/10.1037/pspp0000020

Pennebaker, J. W., Boyd, R. L., Jordan, K., & Blackburn, K. (2015). *The development and psychometric properties of liwc2015* (tech. Rep.).

Pennebaker, J. W., & King, L. A. (1999). Linguistic styles: Language use as an individual difference. *Journal of Personality and Social Psychology*, *77*(6), 1296–1312. https://doi.org/10.1037/0022-3514.77.6.1296

Petrov, S., Das, D., & McDonald, R. (2011). *A universal part-of-speech tagset*. Proceedings 8th international conference on language resources and evaluation (LREC'12) (pp. 2089–2096). European Language Resources Association (ELRA).

Poria, S., Gelbukh, A., Agarwal, B., Cambria, E., & Howard, N. (2013). Common sense knowledge based personality recognition from text. In F. Castro, A. Gelbukh, & M. González (Eds.), *Advances in soft computing and its applications* (pp. 484–496). Springer Berlin Heidelberg. https://doi.org/10.1007/978-3-642-45111-9_42

Preotiuc-Pietro, D., Carpenter, J., Giorgi, S., & Ungar, L. (2016). *Studying the dark triad of personality through twitter behavior*. Proceedings 25th ACM international on conference on information and knowledge management (pp. 761–770). Association for Computing Machinery. https://doi.org/10.1145/2983323.2983822

Preoţiuc-Pietro, D., Eichstaedt, J., Park, G., Sap, M., Smith, L., Tobolsky, V., Schwartz, H. A., & Ungar, L. (2015). *The role of personality, age, and gender in tweeting about mental illness*. Proceedings of the 2nd workshop on computational linguistics and clinical psychology: From linguistic signal to clinical reality (pp. 21–30). Association for Computational Linguistics.

Ramezani, M., Feizi-Derakhshi, M.-R., & Balafar, M.-A. (2022). Text-based automatic personality prediction using kgrat-net: A knowledge graph attention network classifier. *Scientific Reports*, *12*(1), 21453.

Ren, Z., Shen, Q., Diao, X., & Xu, H. (2021). A sentiment-aware deep learning approach for personality detection from text. *Information Processing & Management*, *58*(3), 102532.

Ryan, T., & Xenos, S. (2011). Who uses facebook? An investigation into the relationship between the big five, shyness, narcissism, loneliness, and facebook usage. *Computers in Human Behavior*, *27*(5), 1658–1664. https://doi.org/10.1016/j.chb.2011.02.004

Schwartz, H. A., Eichstaedt, J. C., Kern, M. L., Dziurzynski, L., Ramones, S. M., Agrawal, M., Shah, A., Kosinski, M., Stillwell, D., Seligman, M. E. P., & Ungar, L. H. (2013). Personality, gender, and age in the language of social media: The open-vocabulary approach (T. Preis, Ed.). *PLoS One*, *8*(9), 1–16. https://doi.org/10.1371/journal.pone.0073791

Souri, A., Hosseinpour, S., & Rahmani, A. M. (2018). Personality classification based on profiles of social networks' users and the five-factor model of personality. *Human-centric Computing and Information Sciences*, *8*, 1–15.

Stajner, S., & Yenikent, S. (2020). *A survey of automatic personality detection from texts*. Proceedings 28th international conference on computational linguistics (pp. 6284–6295). International Committee on Computational Linguistics. https://doi.org/10.18653/v1/2020.coling-main.553

Sumner, C., Byers, A., Boochever, R., & Park, G. J. (2012). *Predicting dark triad personality traits from twitter usage and a linguistic analysis of tweets*. Proceedings 2012 11th international conference on machine learning and applications (pp. 386–393). IEEE. https://doi.org/10.1109/ICMLA.2012.218

Sun, X., Cheng, H., Liu, B., Li, J., Chen, H., Xu, G., & Yin, H. (2023). Self-supervised hypergraph representation learning for sociological analysis. *IEEE Transactions on Knowledge and Data Engineering*, *35*(11), 11860–11871. https://doi.org/10.1109/tkde.2023.3235312

Sun, X., Liu, B., Ai, L., Liu, D., Meng, Q., & Cao, J. (2022). In your eyes: Modality disentangling for personality analysis in short video. *IEEE Transactions on Computational Social Systems*, *10*(3), 982–993.

Sun, X., Liu, B., Cao, J., Luo, J., & Shen, X. (2018). *Who am i? Personality detection based on deep learning for texts*. Proceedings 2018 IEEE international conference on communications (ICC). IEEE. https://doi.org/10.1109/icc.2018.8422105

Tang, D., Qin, B., & Liu, T. (2015). *Learning semantic representations of users and products for document level sentiment classification*. Proceedings 53rd annual meeting of the Association for Computational Linguistics and the 7th international joint conference on natural language processing. Association for Computational Linguistics. https://doi.org/10.3115/v1/p15-1098

Tausczik, Y. R., & Pennebaker, J. W. (2009). The psychological meaning of words: Liwc and computerized text analysis methods. *Journal of Language and Social Psychology*, *29*(1), 24–54. https://doi.org/10.1177/0261927x09351676

Uher, J. (2013). Personality psychology: Lexical approaches, assessment methods, and trait concepts reveal only half of the story—Why it is time for a paradigm shift. *Integrative Psychological and Behavioral Science*, *47*, 1–55.

Vaid, S. S., & Harari, G. M. (2021). Who uses what and how often?: Personality predictors of multiplatform social media use among young adults. *Journal of Research in Personality*, *91*, 104005. https://doi.org/10.1016/j.jrp.2020.104005

20448295, 0, Downloaded from https://bpspsychub.onlinelibrary.wiley.com/doi/10.1111/bjop.12755 by University Of Arts London, Wiley Online Library on [21/01/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Wang, Y., Wang, J., & Zhang, X. (2023). In J. Barnes, O. de Clercq, & R. Klinger (Eds.), Proceedings of the 13th workshop on computational approaches to subjectivity, sentiment, & social media analysis *YNU-HPCC atWASSA-2023 shared task 1: Large-scale language model with LoRA fine-tuning for empathy detection and emotion classification* (pp. 526–530). Association for Computational Linguistics. https://doi.org/10.18653/v1/2023.wassa-1.45

Wang, Z., Wu, C.-H., Li, Q.-B., Yan, B., & Zheng, K.-F. (2020). Encoding text information with graph convolutional networks for personality recognition. *Applied Sciences*, *10*(12), 4081.

Wankhade, M., Rao, A. C. S., & Kulkarni, C. (2022). A survey on sentiment analysis methods, applications, and challenges. *Artificial Intelligence Review*, *55*(7), 5731–5780. https://doi.org/10.1007/s10462-022-10144-1

Wen, Z., Cao, J., Yang, Y., Wang, H., Yang, R., & Liu, S. (2023). Desprompt: Personality-descriptive prompt tuning for few-shot personality recognition [art. no. 103422]. *Information Processing and Management*, *60*(5), 103422. https://doi.org/10.1016/j.ipm.2023.103422

Widiger, T. A., & McCabe, G. A. (2020). Personality in clinical psychology. In P. J. Corr, & G. Matthews (Eds.), *The cambridge handbook of personality psychology* (pp. 451–462). Cambridge University Press. https://doi.org/10.1017/9781108264822.042

Wu, Y., Yuan, M., Song, Y., Chen, L., Yuan, C., & Li, Q. (2023). *A two-stage prompt learning method for jointly predicting topic and personality*. 2023 IEEE international conference on intelligence and security informatics (ISI). IEEE. https://doi.org/10.1109/isi58743.2023.10297139

Xue, D., Wu, L., Hong, Z., Guo, S., Gao, L., Wu, Z., Zhong, X., & Sun, J. (2018). Deep learning-based personality recognition from text posts of online social networks. *Applied Intelligence*, *48*(11), 4232–4246. https://doi.org/10.1007/s10489-018-1212-4

Yan, Z., Wang, R., & Sun, X. (2024). What you write represents your personality: A dual knowledge stream graph attention network for personality detection. In F. Wu, et al. (Eds.), *Social media processing. SMP 2023. Communications in computer and information science* (Vol. *1945*). Springer. https://doi.org/10.1007/978-981-99-7596-9_9

Yang, T., Yang, F., Ouyang, H., & Quan, X. (2021). Psycholinguistic tripartite graph network for personality detection. In *Proceedings of the 59th annual meeting of the association for computational linguistics and the 11th international joint conference on natural language processing (Vol. 1: Long papers)* (pp. 4229–4239).

Yarkoni, T. (2010). Personality in 100, 000 words: A large-scale analysis of personality and word use among bloggers. *Journal of Research in Personality*, *44*(3), 363–373. https://doi.org/10.1016/j.jrp.2010.04.001

Zhou, L., Zhang, Z., Zhao, L., & Yang, P. (2022). Attention-based bilstm models for personality recognition from user-generated content. *Information Sciences*, *596*, 460–471. https://doi.org/10.1016/j.ins.2022.03.038

Zhu, Y., Hu, L., Ning, N., Zhang, W., & Wu, B. (2022). A lexical psycholinguistic knowledge-guided graph neural network for interpretable personality detection. *Knowledge-Based Systems*, *249*, 108952.

SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

How to cite this article: Bhandarkar, A., Wilson, R., Swarup, A., Webster, G. D., & Woodard, D. (2024). Bridging minds and machines: Unmasking the limits in text-based automatic personality recognition for enhanced psychology–AI synergy. *British Journal of Psychology*, *00*, 1–23. https://doi.org/10.1111/bjop.12755

20448295, 0, Downloaded from https://bpspsychub.onlinelibrary.wiley.com/doi/10.1111/bjop.12755 by University Of Arts London, Wiley Online Library on [21/01/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License